

ORF 524 — Midterm Exam

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Contents

1	MSE and Point Estimation (50 points)	1
2	MLE Methods: Poisson Distribution (50 points)	2

Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 MSE and Point Estimation (50 points)

Assume $X \sim \mathbb{P}_\theta$ with $\theta \in \Theta$. Let $\hat{\theta}_n = \hat{\theta}_n(X)$ be a square-integrable estimator of θ , not necessarily unbiased. Suppose that $T(X)$ is a sufficient statistic for θ , and define:

$$\tilde{\theta}_n = \tilde{\theta}_n(X) = \mathbb{E}_\theta[\hat{\theta}_n|T(X)].$$

1. Show that $\tilde{\theta}_n$ is an estimator of θ .

Solution: Follows directly by the assumption that $T(X)$ is a sufficient statistic for θ .

2. Show that $\tilde{\theta}_n$ is unbiased (for θ) if and only if $\hat{\theta}_n$ is unbiased (for θ).

Solution: using the law of iterated expectations, $\mathbb{E}_\theta[\tilde{\theta}_n] = \mathbb{E}_\theta[\mathbb{E}_\theta[\hat{\theta}_n|T(X)]] = \mathbb{E}_\theta[\hat{\theta}_n]$, which shows that the two estimators have the same expectation, and the result follows.

3. Show that $\mathbb{V}_\theta[\tilde{\theta}_n] \leq \mathbb{V}_\theta[\hat{\theta}_n]$.

Solution: by the law of total variance,

$$\mathbb{V}_\theta[\hat{\theta}_n] = \mathbb{V}_\theta[\mathbb{E}_\theta[\hat{\theta}_n|T(X)]] + \mathbb{E}_\theta[\mathbb{V}_\theta[\hat{\theta}_n|T(X)]] = \mathbb{V}_\theta[\tilde{\theta}_n] + \mathbb{E}_\theta[\mathbb{V}_\theta[\hat{\theta}_n|T(X)]] \geq \mathbb{V}_\theta[\tilde{\theta}_n],$$

using that $\mathbb{V}_\theta[\hat{\theta}_n|T(X)] \geq 0$.

4. Show that $\mathbb{E}_\theta[(\tilde{\theta}_n - \theta)^2] \leq \mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]$.

Solution: using the previous results,

$$\begin{aligned} \mathbb{E}_\theta[(\tilde{\theta}_n - \theta)^2] &= \mathbb{V}_\theta[\tilde{\theta}_n] + (\mathbb{E}_\theta[\tilde{\theta}_n] - \theta)^2 \\ &= \mathbb{V}_\theta[\tilde{\theta}_n] + (\mathbb{E}_\theta[\hat{\theta}_n] - \theta)^2 \\ &\leq \mathbb{V}_\theta[\hat{\theta}_n] + (\mathbb{E}_\theta[\hat{\theta}_n] - \theta)^2 = \mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]. \end{aligned}$$

5. Determine whether the following statement is true or false:

$$\text{MSE}_\theta[\tilde{\theta}_n] \leq \text{MSE}_\theta[\hat{\theta}_n] \iff \mathbb{E}_\theta[\hat{\theta}_n] = \theta \quad \forall \theta \in \Theta.$$

where $\text{MSE}_\theta[Y] = \mathbb{E}_\theta[(Y - \theta)^2]$ for any square-integrable random variable Y .

Discuss the connections between this question and the Rao-Blackwell Theorem.

Solution: the statement is false. Note that unbiasedness of $\hat{\theta}_n$ is not needed to obtain the inequality of the MSE risk functions, as shown in (1)-(3). The Rao-Blackwell Theorem considers the particular case of unbiased estimators.

Let (U_1, U_2) be a random sample from the distribution $\text{Uniform}(0, \theta)$, where $\Theta = (0, \infty)$. Recall that $T(X) = \max(U_1, U_2)$ is a complete minimal sufficient statistics for θ .

6. Show that

$$F_{U_1|\max(U_1, U_2)}(u|v) = \begin{cases} 0 & \text{if } u \leq 0 \\ \frac{u}{2v} & \text{if } 0 < u < v < \theta \\ 1 & \text{if } v \leq u \end{cases}$$

Solution:

$$\begin{aligned} F_{U_1|\max(U_1, U_2)}(u|v) &= \mathbb{P}[U_1 \leq u | \max(U_1, U_2) = v] \\ &= \mathbb{P}[U_1 \leq u, U_1 \leq U_2 | \max(U_1, U_2) = v] + \mathbb{P}[U_1 \leq u, U_1 > U_2 | \max(U_1, U_2) = v] \\ &= \mathbb{P}[U_1 \leq u | U_1 \leq U_2, \max(U_1, U_2) = v] \cdot \mathbb{P}[U_1 \leq U_2 | \max(U_1, U_2) = v] \\ &\quad + \mathbb{P}[U_1 \leq u | U_1 > U_2, \max(U_1, U_2) = v] \cdot \mathbb{P}[U_1 > U_2 | \max(U_1, U_2) = v] \\ &= \mathbb{P}[U_1 \leq u | U_1 \leq U_2, U_2 = v] \cdot \frac{1}{2} + \mathbb{P}[U_1 \leq u | U_1 > U_2, U_1 = v] \cdot \frac{1}{2}, \end{aligned}$$

and the result follows.

7. Show that $\mathbb{E}_\theta[U_1 | \max(U_1, U_2)] = \frac{3}{4} \max(U_1, U_2)$.

Solution:

$$\mathbb{E}_\theta[U_1 | \max(U_1, U_2) = v] = \int u F_{U_1|\max(U_1, U_2)}(du|v) = \left(\int_0^v u \frac{1}{2v} du + \frac{1}{2}v \right) = \frac{3}{4}v.$$

8. Show that $\hat{\theta}_n = U_1 + U_2$ is an unbiased estimator of θ .

Solution: $\mathbb{E}_\theta[\hat{\theta}_n] = \mathbb{E}_\theta[U_1 + U_2] = \frac{\theta}{2} + \frac{\theta}{2} = \theta$ and hence $\hat{\theta}_n$ is unbiased for θ .

9. Show that $\tilde{\theta}_n = \frac{3}{2} \max(U_1, U_2)$, and that it is also an unbiased estimator of θ .

Solution:

$$\tilde{\theta}_n = \mathbb{E}_\theta[\hat{\theta}_n | T(X)] = \mathbb{E}_\theta[U_1 + U_2 | \max(U_1, U_2)] = 2\mathbb{E}_\theta[U_1 | \max(U_1, U_2)] = \frac{3}{2} \max(U_1, U_2).$$

Furthermore, $\mathbb{E}_\theta[\tilde{\theta}_n] = \mathbb{E}_\theta[\mathbb{E}_\theta[\hat{\theta}_n | T(X)]] = \mathbb{E}_\theta[\hat{\theta}_n] = \theta$ and hence $\tilde{\theta}_n$ is also unbiased.

10. Let $\check{\theta}_c = c \max(U_1, U_2)$, for $c \in \mathbb{R}$. Show that

$$\frac{\text{MSE}_\theta[\check{\theta}]}{\text{MSE}_\theta[\hat{\theta}]} = \frac{3}{4} < 1, \quad \text{and} \quad \frac{\text{MSE}_\theta[\check{\theta}_c]}{\text{MSE}_\theta[\tilde{\theta}]} = \frac{8}{9} < 1,$$

for appropriate choice of c .

Explain in which sense $\tilde{\theta}$ is an optimal point estimator of θ , and when $\check{\theta}_c$ may be regarded as a better point estimator than $\tilde{\theta}$ (for appropriate choice of c).

Solution: First, recall we have

$$\begin{aligned}\mathbb{E}_\theta[U_1 + U_2] &= \theta, & \mathbb{V}_\theta[U_1 + U_2] &= \frac{1}{6}\theta^2, \\ \mathbb{E}_\theta[\max(U_1, U_2)] &= \frac{2}{3}\theta, & \mathbb{V}_\theta[\max(U_1, U_2)] &= \frac{1}{18}\theta^2\end{aligned}$$

where the last result follows by

$$\mathbb{V}_\theta[\max(U_1, U_2)] = \mathbb{E}_\theta[\max(U_1, U_2)^2] - (\mathbb{E}_\theta[\max(U_1, U_2)])^2 = \frac{1}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{1}{18}\theta^2.$$

Then,

$$\frac{\text{MSE}_\theta[\tilde{\theta}]}{\text{MSE}_\theta[\hat{\theta}]} = \frac{\mathbb{V}_\theta[\tilde{\theta}]}{\mathbb{V}_\theta[\hat{\theta}]} = \frac{\frac{9}{4}\mathbb{V}_\theta[\max(U_1, U_2)]}{\mathbb{V}_\theta[U_1 + U_2]} = \frac{\frac{9}{4}\frac{1}{18}\theta^2}{\frac{1}{6}\theta^2} = \frac{3}{4} < 1.$$

Furthermore,

$$\begin{aligned}\frac{\text{MSE}_\theta[\check{\theta}]}{\text{MSE}_\theta[\tilde{\theta}]} &= \frac{\mathbb{V}_\theta[\check{\theta}] + (\mathbb{E}_\theta[\check{\theta}] - \theta)^2}{\mathbb{V}_\theta[\tilde{\theta}]} \\ &= \frac{\mathfrak{c}^2\mathbb{V}_\theta[\max(U_1, U_2)] + (\mathfrak{c}\mathbb{E}_\theta[\max(U_1, U_2)] - \theta)^2}{\frac{9}{4}\mathbb{V}_\theta[\max(U_1, U_2)]} \\ &= \frac{\mathfrak{c}^2\frac{1}{18}\theta^2 + (\mathfrak{c}\frac{2}{3} - 1)^2\theta^2}{\frac{1}{8}\theta^2} = \frac{4}{9}\mathfrak{c}^2 + 8\left(\mathfrak{c}\frac{2}{3} - 1\right)^2\end{aligned}$$

and minimizing over $\mathfrak{c}$ we obtain the optimal $\mathfrak{c}^\star = \frac{4}{3}$, which gives

$$\frac{4}{9}(\mathfrak{c}^\star)^2 + 8\left(\mathfrak{c}^\star\frac{2}{3} - 1\right)^2 = \frac{8}{9} < 1.$$

The estimator $\tilde{\theta} = \frac{3}{2}\max(U_1, U_2)$ is the Uniform Minimum Variance Estimator among unbiased estimators. However, the (biased for θ) estimator $\check{\theta}_{\mathfrak{c}^\star} = \frac{4}{3}\max(U_1, U_2)$ has smaller mean square error than $\tilde{\theta}$ because it has less variance but more bias. Thus $\check{\theta}_{\mathfrak{c}^\star}$ is a better estimator for θ than $\tilde{\theta}$ from a MSE perspective, but not from an efficiency (minimum variance among unbiased estimators) perspective.

2 MLE Methods: Poisson Distribution (50 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta = \text{Poisson}(\theta)$, with $\theta \in (0, \infty)$.

1. Find the Maximum Likelihood (ML) estimator of θ , denoted by $\hat{\theta}_{\text{ML}}$.

Solution: We have $f_{X_i}(x; \theta) = \frac{\theta^x}{x!} \exp(-\theta) \mathbf{1}(x \in \mathbb{Z}_+)$, with $\theta \in \Theta = \mathbb{R}_+$. The individual log-likelihood is:

$$l_i(\theta; x) = x \log \theta - \log x! - \theta + \log(\mathbf{1}(x \in \mathbb{Z}_+)),$$

and hence

$$\frac{\partial}{\partial \theta} l_i(\theta; x) = \frac{x}{\theta} - 1, \quad \frac{\partial^2}{\partial \theta^2} l_i(\theta; x) = -\frac{x}{\theta^2}.$$

Thus,

$$l_n(\theta) = \sum_{i=1}^n x_i \log \theta - n\theta - \sum_{i=1}^n \log x_i! + \sum_{i=1}^n \log(\mathbf{1}(x_i \in \mathbb{Z}_+))$$

If $\sum_{i=1}^n x_i = 0$, the function is maximized at 0 so $\hat{\theta}_{\text{MLE}} = 0$. On the other hand, if $\sum_{i=1}^n x_i > 0$ we can use the FOC and SOC:

$$\begin{aligned} \frac{\partial}{\partial \theta} l_n(\theta) &= \frac{\sum_{i=1}^n x_i}{\theta} - n \Rightarrow \hat{\theta}_{\text{ML}} = \bar{X}_n \\ \frac{\partial^2}{\partial \theta^2} l_n(\theta) &= -\frac{\sum_{i=1}^n x_i}{\theta^2} < 0 \end{aligned}$$

Since the function is continuous and goes to $-\infty$ as $\theta \rightarrow 0$, there are no boundary solutions in this case.

2. Show that $\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta) \rightsquigarrow \text{N}(0, \theta)$.

Solution: Since $\hat{\theta}_{\text{ML}} = \bar{X}_n$, and $\mathbb{E}[X_i] = \theta = \mathbb{V}[X_i]$, the result follows directly by the CLT.

3. Is $\hat{\theta}_{\text{ML}}$ asymptotically efficient for θ ?

Solution: We have

$$\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} l_i(\theta; X_i) \right] = \frac{\mathbb{E}[X_i]}{\theta^2} = -\frac{1}{\theta}.$$

Thus, the CRLB is $-\left(n \mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} l_i(\theta; X_i) \right]\right)^{-1} = \theta/n = \mathbb{V}[\bar{X}_n]$. Thus, the estimator is (UMVU and hence) asymptotically efficient.

4. Propose a feasible, asymptotically valid $100(1 - \alpha)\%$ confidence interval estimator for θ , denoted by $\text{CI}_{1-\alpha}(\theta)$, for $\alpha \in [0, 1]$.

Solution: An $(1 - \alpha)$ -confidence level CI is

$$\text{CI}_{1-\alpha}(\theta) = \left[\bar{X}_n - \Phi^{-1}(1 - \alpha/2) \sqrt{\frac{\bar{X}_n}{n}}, \bar{X}_n + \Phi^{-1}(1 - \alpha/2) \sqrt{\frac{\bar{X}_n}{n}} \right],$$

where $\Phi(x)$ denotes the cumulative distribution function of the standard Gaussian distribution.

Consider now the parameter $\beta = \beta(\theta) = \mathbb{P}_\theta[X_i = 0]$.

5. Find the ML estimator of β , denoted by $\hat{\beta}_{\text{ML}}$.

Solution: Since $\beta = \beta(\theta) = \mathbb{P}_\theta[X_i = 0] = \exp(-\theta)$, the MLE is $\hat{\beta}_{\text{ML}} = \exp(-\hat{\theta}_{\text{ML}})$

6. Show that $\sqrt{n}(\hat{\beta}_{\text{ML}} - \beta) \rightsquigarrow \text{N}(0, \theta \exp(-2\theta))$.

Solution: Since $x \mapsto g(x) = \exp(-x)$ is differentiable with $\frac{d}{dx}g(x) = -\exp(-x) \neq 0$, the delta method gives

$$\sqrt{n}(\hat{\beta}_{\text{ML}} - \beta) = \sqrt{n}(g(\hat{\theta}_{\text{ML}}) - g(\theta)) \rightsquigarrow \left(\frac{d}{dx}g(\theta) \right) \text{N}(0, \theta) =_{\text{d}} \text{N}(0, \theta \exp(-2\theta))$$

7. Is $\hat{\beta}_{\text{ML}}$ asymptotically efficient for β ?

Solution: Yes, because it is asymptotically normal with asymptotic variance $g'(\theta)^2 I(\theta)^{-1} = e^{-2\theta} \theta / n$. Note it is also UMVUE for each n !

8. Propose a feasible, asymptotically valid $100(1 - \alpha)\%$ confidence interval estimator for β , denoted by $\text{CI}_{1-\alpha}(\beta)$.

Solution: We have shown that $\sqrt{n}(e^{-\bar{X}_n} - \beta) \rightsquigarrow \text{N}(0, \theta e^{-2\theta})$. Combining this with $e^{\bar{X}_n} / \sqrt{\bar{X}_n} \rightarrow_p e^\theta / \sqrt{\theta}$, we have by Slutsky's lemma that

$$\frac{\sqrt{n}(1 - e^{\bar{X}_n} \beta)}{\sqrt{\bar{X}_n}} \rightsquigarrow \text{N}(0, 1).$$

Since the distribution function of $\text{N}(0, 1)$ is continuous, this implies

$$\mathbb{P}_\beta \left\{ -c_\alpha \leq \frac{\sqrt{n}(1 - e^{\bar{X}_n} \beta)}{\sqrt{\bar{X}_n}} \leq c_\alpha \right\} \rightarrow \Phi(c_\alpha) - \Phi(-c_\alpha) = 2\Phi(c_\alpha) - 1.$$

Rewriting and putting $c_\alpha = \Phi^{-1}(1 - \alpha/2)$, we obtain the confidence interval

$$\left[-\Phi^{-1} \left(1 - \frac{\alpha}{2} \right) e^{-\bar{X}_n} \sqrt{\frac{\bar{X}_n}{n}} + e^{-\bar{X}_n}, \Phi^{-1} \left(1 - \frac{\alpha}{2} \right) e^{-\bar{X}_n} \sqrt{\frac{\bar{X}_n}{n}} + e^{-\bar{X}_n} \right].$$

Consider the following one-sided hypothesis testing problem:

$$\text{H}_0 : \beta \geq \beta_0 \quad \text{vs.} \quad \text{H}_1 : \beta < \beta_0, \quad \text{for some fixed } \beta_0 \in (0, 1).$$

9. Propose a a feasible, asymptotically α -level valid hypothesis test.

Solution: We will find a test of the form

$$\text{reject } H_0 \iff \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta_0}{\sqrt{\bar{X}_n}} \leq c_\alpha.$$

If $\beta \geq \beta_0$, we have

$$\begin{aligned} \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta_0}{\sqrt{\bar{X}_n}} \leq c_\alpha \right\} &= \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} + \sqrt{n} \frac{e^{\bar{X}_n} (\beta - \beta_0)}{\sqrt{\bar{X}_n}} \leq c_\alpha \right\} \\ &\leq \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq c_\alpha \right\} \rightarrow \Phi(c_\alpha). \end{aligned}$$

Putting $c_\alpha = \Phi^{-1}(\alpha)$ gives an asymptotically α -level feasible test.

10. Show that the proposed hypothesis test is consistent.

Solution: If $\beta < \beta_0$, we have

$$\mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta_0}{\sqrt{\bar{X}_n}} \leq c_\alpha \right\} = \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq c_\alpha + \sqrt{n} \frac{e^{\bar{X}_n} (\beta_0 - \beta)}{\sqrt{\bar{X}_n}} \right\}.$$

Note that since $e^{\bar{X}_n} (\beta_0 - \beta) / \sqrt{\bar{X}_n}$ converges in probability to a positive constant, the quantity $\xi_n := c_\alpha + \sqrt{n} e^{\bar{X}_n} (\beta_0 - \beta) / \sqrt{\bar{X}_n}$ is greater than any fixed $M > 0$ with probability approaching one. Then

$$\begin{aligned} \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq \xi_n \right\} &\geq \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq \xi_n, \xi_n > M \right\} \\ &\geq \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq M, \xi_n > M \right\} \\ &\geq \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq M \right\} + \mathbb{P}_\beta \{ \xi_n > M \} - 1 \\ &= \mathbb{P}_\beta \left\{ \sqrt{n} \frac{1 - e^{\bar{X}_n} \beta}{\sqrt{\bar{X}_n}} \leq M \right\} + o(1) = \Phi(M) + o(1). \end{aligned}$$

Since $M > 0$ is arbitrary, we have shown that the probability on the left of this chain is eventually (as $n \rightarrow \infty$) greater than any number less than one, which means that it tends to one, proving consistency.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$